

Network Concomitance:

An AI and Network Science Approach to Theory Building

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Abstract. This paper advances the concept of **network concomitance**, arguing that social, semantic, and interaction networks are inherently co-evolving systems that must be modeled jointly to support theory building. Drawing on recent advances in artificial intelligence and network science, it introduces an integrated framework for generating and evaluating synthetic social media environments using large language models (LLMs). Central to this approach is the GhostField system, which combines scenario design, persona construction, and iterative validation to produce realistic synthetic data that reflects both structural and behavioral dynamics.

The study highlights limitations in early-generation synthetic networks, including weak community structure, excessive centralization, and lack of narrative connectivity across messages. These shortcomings illustrate the difficulty of reproducing the cognitive and social constraints that shape real-world interactions. Second-generation models show substantial improvements, including enhanced modularity, realistic linking patterns, and stronger alignment between actors, content, and network structure. However, challenges remain in achieving semantic coherence and balancing interpretability with complexity.

To address these issues, the paper emphasizes the incorporation of social cognition, including attention, priming, and schema activation, as well as behavioral and environmental constraints such as time, co-presence, and group membership. By modeling actors with distinct identities, roles, and linguistic styles, the framework enables more realistic simulations of communication and influence. Ultimately, the paper argues that robust synthetic data must capture not only network structure, but the dynamic interplay among actors, meaning, and interaction.

Keywords: Network Science, Large Language Models, Synthetic Social Media Data

1 Introduction

In an era defined by rapidly evolving information ecosystems, the ability to generate realistic synthetic social media and media data is no longer a luxury — it is a necessity. As misinformation campaigns grow in sophistication and scale, researchers and practitioners require environments where detection methods can be developed, stress-tested, and refined without endangering real populations or relying on incomplete data. Synthetic data offers precisely this capability, enabling the creation of rich, controlled, and ethically safe simulations of information flows, narrative competition, and coordinated influence. Beyond detection, these environments are indispensable for planning and rehearsing crisis response, allowing decision-makers to anticipate how information spreads, how publics react, and how interventions might succeed or fail under pressure. They also provide a critical foundation for training analysts, preparing them to recognize emerging patterns of manipulation and to act with speed and confidence. Finally, synthetic ecosystems serve as rigorous testbeds for evaluating tools and technologies, ensuring that systems designed to monitor, interpret, and respond to information dynamics are robust before deployment. By capturing the intertwined evolution of actors, content, and interactions, this approach does not merely replicate data—it creates a space for discovery, foresight, and resilience in the face of complex informational threats.

The fundamental feature of social society that synthetic data must capture is network concomitance. Networks are concomitant in that they naturally co-occur and influence one another. Social relationships shape information exposure, and knowledge structures influence whom individuals interact with. This reciprocal dynamic creates layered systems in which social, semantic, and interaction networks evolve together. Traditional theories such as constructivism [1,2] when realized as an AB-DN-M [3] approximate these dynamics, but they often fail to produce fully integrated semantic and knowledge representations where meaning is grounded in realistic language usage. Incorporating LLM-based agents provides an opportunity to bridge this gap.

The integration of artificial intelligence and network science has opened new pathways for understanding complex social systems. This paper examines how simulation can be used to generate synthetic data that exhibits network concomitance, defined as the simultaneous and interdependent evolution of social (who to whom), semantic (idea to idea), and knowledge (who to idea) networks. The discussion outlines how synthetic data generation, generated by a combination of agent-based dynamic-network simulation (AB-DN-Ms) and large language models (LLMs), can be used to construct realistic network environments for theory building and simulation, the limits of such simulations, and the future directions needed.

2 Synthetic Data Generation and GhostField

The GhostField framework [4] contains a pipeline for producing synthetic data that mimics real-world social systems. It integrates scenario descriptions, persona modeling, and synthetic media creation into a simulation pipeline composed out of an AB-DN-M and an LLM. The AB-DN-M generates co-evolving networks and the LLM generates content such as social media posts and news articles. Ghostfield contains a simulation and validation pipeline, as shown in Figure 1. AESOP is used by the scenario designer to lay out the story line for the scenario – key events, issues, actors, and groups [5]. AESOP automatically generates out news stories, a scenario description containing personae profiles so that SynSm can generate out the synthetic social media posts. SynSm generates these posts along with meta-data in the same format as the platform. This can then be read into and analyzed using Artemis and evaluated using MOMUS. Artemis is the analytic engine that includes ORA[6] and other analytic tools that support analyzing social media data. The MOMUS system within GhostField assesses synthetic social media data statistically to ensure realism on over 200 indicators and that it matches the scenario. MyMuse is a tool that can be used by participants in a live exercise to contribute their own changes to the simulation for the next day’s data. This iterative generate-validate process enables the refinement of synthetic data so that it better reflects behavioral and structural patterns observed in real environments.

The SynSm system has theoretical processes built in it. The agent-based dynamic-network component, as it is predicated on constructivism theory, has a social influence[7] and homophily processes[8]. Agents are more likely to interact with those with whom they share more knowledge, as agents interact they share knowledge which influences whom they will interact with, and agents will based on their knowledge and understanding of the opinions of those they interact with gradually change their opinions. The LLM component provides the ability to represent knowledge and opinion in complex phrases and also provides an implicit mimicry mechanism – that is, LLMs by their nature tend to mimic other LLMs they interact with as they “learn” in a way that allows them to reconstruct.

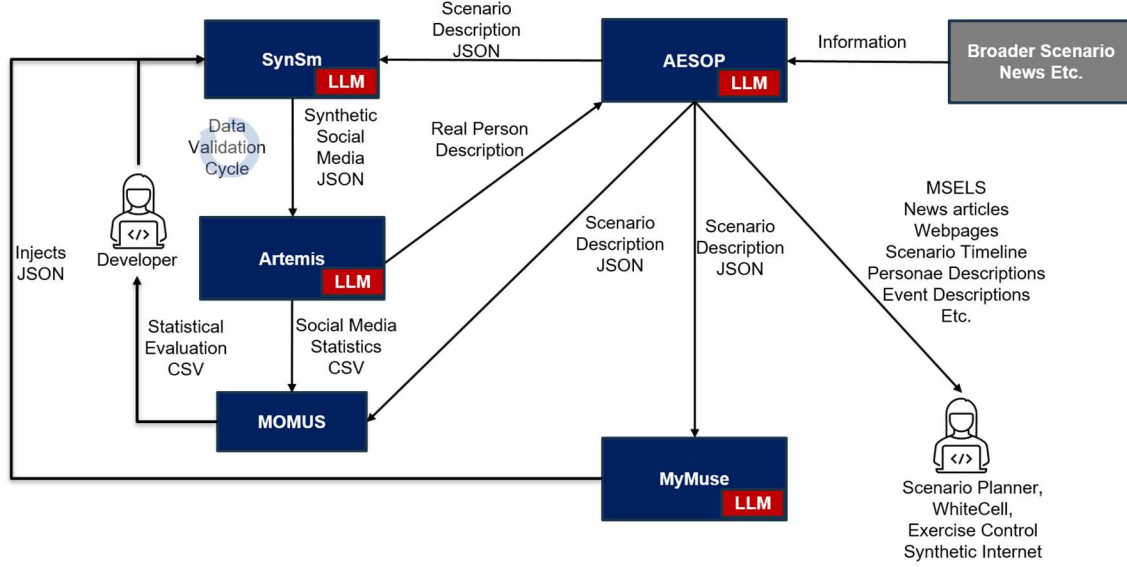


Fig. 1. GhostField Pipeline.

3 General Characteristics of Synthetic Data

Synthetic data generated through LLMs can resemble real user-generated content to humans. That is, it is easy to get most LLMs to generate social media posts, e.g., tweets, that when the random person reads them, they will think they are a tweet by a human. Indeed, this is one of the reasons that people are becoming increasingly skeptical that it will be possible to identify tweets from bots, including emotionally charged, persuasive, and reactive language [9]. However, such systems are producing unrealistic patterns, including overly positive tone, lack of narrative cohesion, and implausible interaction structures [10], factors made worse by the inherent bias due to the corpora on which these LLMs are trained and well-meaning but often distorting limitations placed on the LLMs by the developers [11]. These limitations highlight several things: First, humans simply reading posts are not good judges of the quality of synthetic data – rather a more empirical statistical assessment is needed. Second, given differences in training and in brakes – there are likely to be differences in the quality of the synthetic data produced by different LLMs. And third, there is a need to embed behavioral and cognitive realism into the synthetic generation process which may require new theory.

To address these limitations, the following approach was taken. A review of the synthetic data and of multiple real world social network data sets was done. This was done in a series of waves where in each wave synthetic data was generated, evaluated, then changes were made to GhostField (and so the underlying theory). The expectation is that through this process, our understanding of the precepts of concomitance will be discovered and as our theory improves so to will the model and the resultant synthetic data.

First, the realism of the synthetic data is assessed using qualitative and quantitative assessments rather than using public opinion. The key feature was that all data sets were meta-networks containing the following features: 1) multi-day data, 2) there were multiple interlinked social networks including e.g. reposting, mentioning, replying, 3) there were multiple semantic networks including hashtag-co-occurrence, url co-occurrence, word co-occurrence, 4) there was a message-to-message network, 5) there were multiple knowledge networks e.g., agent-to-hashtag, agent-to-url, agent-to-word, 6) the agents were of diverse types specifically individuals, bots, news agencies, government actors and corporations, and 7) the nodes whether agents, hashtags, messages, etc. had attributes. A set of stylized facts about human social networks and semantic networks was created and used to orient the search for differences – these are shown in Table 1. Each generation of synthetic data was run through ORA/Artemis to calculate general network

statistics and to visualize the networks. Visual inspection of the differences in the synthetic data and real data were identified and categorized. This qualitative assessment helped to inform the judgement of whether the data exhibited the stylized facts. In addition, for each generation of synthetic data, MOMUS was run on all data. The results included empirical feedback on over 200 dimensions. These dimensions cover three broad categories: network structures, semantics and syntax, and fit to the scenario. This provided guidance for each dimension as to whether the synthetic data was on average too low or too high. This combined assessment approach was taken because the number of possible differences between synthetic and real was potentially infinite, there are significant differences between real-world networks, and for the purposes of training people about the nature of the information environment and to test out new tools it was not necessary to have 100% accurate data just synthetic data that was reasonably accurate on key features.

Table 1: Stylized Network Facts			
Actors & Posts	Social Network	Semantic Network	Knowledge Network
Multiple types of actors	Heavy tailed degree distribution (especially for reposting)	Strong modular/topic structure	Dense, highly structured projections
Extreme inequality in activity and attention	High triadic clustering within communities	Often larger and denser than social networks	Heavy tailed distributions for people and ideas
Highly skewed diffusion	Dense core and sparse periphery	Zipf's law & power law degree distribution	
Event based burstiness	Homophily (those who connect share attributes)		

Second, we compared multiple LLMs. It is worth noting that GhostField is designed so that the LLMs can be easily swapped. Two of the LLMs that did the best for these tasks were Ollama and OpenAI – see Fig. 2. Two dimensions were examined to compare the LLMs – the overall realism and exactness. Realism is assessed on several hundred indicators the extent to which the statistical mean for each measure based on the synthetic data lies within the distribution for that measure in the real data. Exactness uses several dozen measures to assess whether the synthetic data matches the pre-scribed levels of various indicators as required by the scenario (e.g., number of posts, number of actors...). Model evaluation indicates that the choice of LLM significantly impacts output quality. These results confirm that the LLM does make a difference and that this difference on the metric being matched but are similar across platforms. Metrics such as realism and exactness reveal differences between LLM models in their ability to reproduce structural and semantic features. While improvements in exactness were achieved through iterative refinement, challenges remain in aligning semantic coherence with structural realism that are independent of the LLM used. Due to these challenges, and the fact that the cross-platform effects are bigger than the cross LLM effects, for the rest of the study the focus will be on Ollama results and synthetic X data.

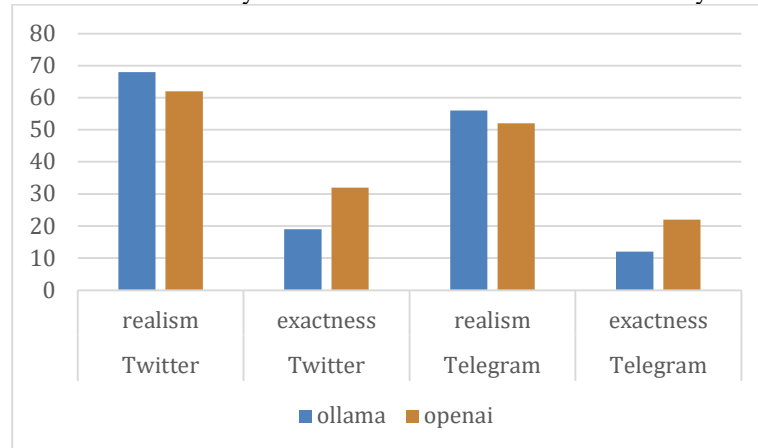


Fig 2. Effect of different LLMs on the realism and exactness of the synthetic data.

Third we used the data from the assessment of each wave of data used it to define the set of general characteristics that the synthetic data must match to get the desired level of behavioral and cognitive realism. That is, to identify where there are weak spots in our current theory. This multi-step process and its effectiveness is illustrated using data from four waves.

3.1 Generation 1 Synthetic Networks

First-generation synthetic networks reveal significant limitations.¹ At the user level, social network structures tend to be overly centralized, dominated by a few high-degree actors. Other actors tend to be isolates. The social network lacks a community structure and fails to capture realistic clustering patterns. At the message level, networks exhibit modularity but lack inter-group connections and narrative linkage. You can't, for example, follow hashtags from one actor to the next such that group's can be discerned by what hashtags they use. – i.e., the knowledge network appears to be random. The hashtag to semantic network, based on the hashtag-to-hashtag network, is a single bramble and does not have a community structure that breaks into distinct narratives. Rather, the same hashtags are repeated indiscriminately. These shortcomings prevent accurate modeling of information diffusion and group dynamics.

The posts often read like things that could have appeared in social media. However, they contained too little emotional language, rarely mentioned other users, and frequently contained prompt leakages. In addition, many posts were simply real-world posts with the names and dates changed and so did not match the scenario. Overall they failed on the vast majority of MOMUS dimensions.

Key changes made before generating the next version of synthetic data were the following: 1) producers of LLMs released new versions, 2) constraints on prompts to reduce prompt leakage were introduced, 3) LLMs were instructed to include hashtags in social media posts (probabilistically), 4) Synthetic generators were given an increased likelihood of retweets and quotes, 5) parameter settings for the use of emotional language were increased, and illustrative emotional terms were provided, and 6) complete synthesis rather than a modify approach was used. Most of these changes were to the parameters in the simulation engines. Essentially, all changes at this stage were in choice of LLM, and in parameters and prompts. The data was so far from realistic that limitations to the theory could not be assessed.

3.2 Generation 2 Improvements

Second-generation synthetic networks demonstrate notable improvements – see Figure 2. They exhibit realistic linking distributions, stronger community structures, and better inter-group connections in both the social and the semantic networks. At this point it is possible to identify communities of actors and within each find opinion leaders. Moreover, the retweet network began to have the typical star like structure typical of real data; however, the centers of the stars were rarely newspapers – whereas in real data they frequently are newspapers. It is possible to find communities of ideas in the semantic network and so identify different narratives. However, simplifications remain, such as constrained issue diversity and limited hashtag usage. The narratives that are identifiable are trivial – and the posts about them highly repetitive. The scarcity of hashtags meant that many narratives were isolated – and not connected into the overall discussion. The knowledge network did not resemble real world knowledge networks - too few hashtags and overuse of those hashtags. Consequently, while actors could be divided into communities – all communities appeared to be talking about the same things – from the perspective of hashtags. These constraints improve interpretability but still reduce realism in representing complex societal discourse.

The posts often read like things that could have appeared in social media. However, they still contained too little emotional language, rarely mentioned other users, and frequently contained prompt leakages. Posts were not simply real-world posts with the names and dates changed – but were new scenario related posts. There was little diversity in content, Too many of these posts sounded official and there was little branching from one message to the next.

¹ For additional analysis of generation 1 and 2 of data see paper presented at SBP-Brims 2025 by Morgan et al. [12].

Essentially, generation 2 showed that parameter adjustments were insufficient. More specifically, altering prompts and parameters did lead to individual networks with better structural properties. What they did not help was the concomitance – thus changes or behavior in one network was not reflected in behavior in other networks. Moreover, they did not help with the “hidden features” of the networks – the non-structural features of the nodes. Thus, although there was an increase in the realism of the data, there were still a large number of limitations. Two more rounds of adjustments to the synthetic data based only on parameter adjustments in the simulator led to minor improvements but the major issues identified remained. This suggested that major factors, sub-models, theoretical relations, etc. were not being adequately captured.

4 The Nature of Network Concomitance

The results of parameter and prompt adjustments were that the synthetic data was better qualitatively on the social network stylized facts than the semantic and better on the semantic than the knowledge network facts. In terms of the actors and posts, the stylized facts were not met. Three factors stood out as potential contributors to the lack of concomitance – the lack of multiple actor types, the lack of burstiness of messages around events, and randomness of ties between actors and issues.

Different types of actors – particularly news, government, organizations, individuals and bots are known to behave differently in social media with different activity levels, different concerns on issues, different communication styles and so receive different levels of attention from other actors. Not only do agents of different types act differently but there are also large differences in the number of agents of each type. For example, the largest class type are individuals – many of whom rarely post but none-the-less use the social media to get information. This suggests that the basic theory should account for actor types – but not through the “matthew effect” but more fundamentally in terms of roles – or as we refer to them actor types. Having different actor tiers with different communication properties and different population sizes should provide the heterogeneity needed to generate complex behavior with the right structural and actor level features. Mimicry and social influence – which were already in the simulator should be able to expand related activity once given a scaffold based on types. Moreover, this approach would support ease of creating the artificial society on the part of the designer as now they could focus on creating only a few foreground actors, a few of each type, and let the rest – the “unsung majority” be created automatically by the simulator as background actors.

Social media is notoriously bursty – with most events receiving no attention and others getting high level of attention that then reduces exponentially. None of the basic theoretical processes in Ghostfield naturally lead to bursts; i.e., none of social influence, homophily, or mimicry alter behavior based on emotions or the criticality of an event, both factors often associated with bursts. Note, simply increasing the parameters for the amount of happy, dismay, anger, and fear emotions displayed does not lead to burstiness, it simply makes more posts emotional but there is nothing inherent in social influence, homophily or mimicry that leads to emotional posts being disproportionately responded to or reposted or agents in emotional states being more active. Alternatively, one could adjust the theory by providing tiers of events, a salience to events, and make agents more likely to discuss salient events longer and more actively. Another alternative would be to add an emotion model such that agents in heightened emotional states are more prone to act – and so send more messages. Through the processes already in the GhostField simulator either or both of these changes should produce bursts.

In the real world the knowledge network is not random. Most people do not talk randomly about things. Rather, they have a viewpoint, an opinion, on an issue and they espouse that view. They can and do change opinions, due to social influence. Social influence alone, though, was insufficient. While the agents were changing opinions, it was a chaotic change. All agents talked about all issues, the statements they made were fairly random, they would shift opinions early and then fixate even on opinions that were at odds with other of their attributes. For example, an Apple employee might start extolling the virtues of Google, an agent allergic to apples might start talking about a taste test they were doing on apples. A possible solution was to put agents into “issue” groups, and to link these issues to events. This way when an event happened any relevant issues would start being talked about by those in the group and the agents

would change their opinions based on social influence, but due to the issue groups their social networks and semantic networks would have greater initial concomitance.

For generation three – the following changes were thus made. Agent classes and issue groups were introduced, and these agents had an emotional state such that a heightened state could as relevant increase activity. Further a RAG based model for generating background agents was introduced so that heterogeneity in opinions by, e.g., country, could be added systematically [13]. Finally checks on personae creation were added to ensure that the personae were internally consistent; i.e., no 5 year olds with college degrees in physics working as cat groomers appeared – instead age, gender, occupation, etc. were aligned.

The result was synthetic data that fit the stylized facts somewhat better. News agencies were more likely to be the most reposted, there was better alignment between agent attributes and the issues they discussed, and there was some burstiness to the data. However, as shown in Figure 3, the data was still poor on most of the MOMUS indicators. The changes were sufficiently promising that they suggested that the idea of types was valuable and should be extended to issues and events as well. The results also suggested that the idea of salience and groups needed to be grounded in stronger theory.

Further assessment led to the realization that many of the data limitations fell into three categories: prompts not being optimized, social cognition not being adequately modeled, and technical features of the platforms and physical human limitations not being captured. These are summarized in Table 2 – where the identified problem and then a potential solution is shown. For each of these solutions those that were implemented between Generation 3 and 4 of the synthetic data are shown in italics.

Table 2: Summary of Limitations by Category			
	Parameters & Prompts	Social Cognition	Tech & Physical Limits
Agents & Posts Problem Possible Solution	Too many from background actors <i>Reduce number of background actors</i>	Frequency doesn't match similar real actors Set baselines by class Lack of role heterogeneity Employ event archetypes Employ issue archetypes Employ actor archetypes – recruiter, adversary ...	Uniform pattern of life Add time of day, holidays, amt time active
Social Network Problem Possible Solution	Small stars <i>Increase number of bots</i>	Community structure inappropriate Employ nominal groups <i>Employ issue groups</i>	Actor and network mismatch Add joining constraints Add time active
Semantic Network Problem Possible Solution	Mimicry too high Employ linguistic styles Low emotional language <i>Prompt for emotion style</i> Zipfs law not strong enough <i>Add RAGS or longer prompts for key issues & events</i>	Stance via label propagation fails Use a consistent direction in setting stance by group Set adversaries on opposing narratives <i>Set pro-con-neutral by narrative</i> <i>Attach narratives to issues</i> <i>Attach issues to events</i> <i>Set issues by group</i>	Words of interest don't exist Add workarounds for key platform policies on use of abusive language, post length ...
Knowledge Network Problem Possible Solution	All agents talk on all issues and from all narratives <i>Limit agents to only discussing salient issues</i>	Distribution not right heavy tailed for agents or ideas Set narratives by multiple types of groups	Agents learn new issues too fast and change opinions too fast Use salience and time active to naturally reduce learning rate

	Use agent groups to assign agents to issue classes		
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After making these changes we do see some improvement in the data – both better fit to the stylized facts and better numerical fit on the dimensions used by MOMUS. Figure 3 shows how these changes impacted the synthetic data overall in moving from generation 3 to 4. As can be seen overall, the change in realism was negligible; however, the synthetic data now followed the scenario better and so appeared to be much better data.

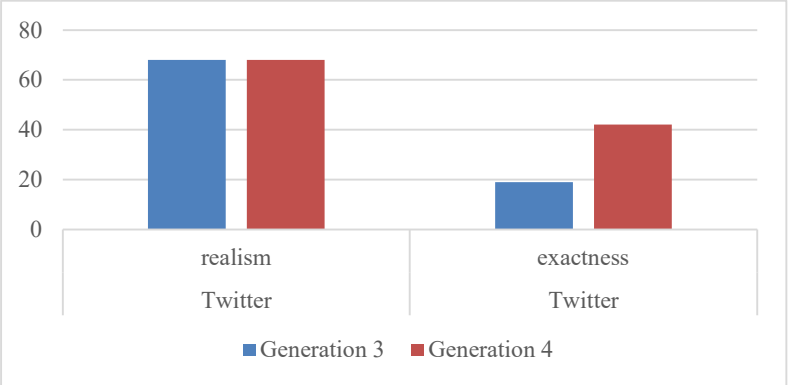


Figure 3. Impact of theoretical adjustments on synthetic data

In Figure 4, a more detailed comparison of generation 3 and 4 data is provided. On every category of metric – except network – the changes made resulted in better data. Ironically, the changes did lead to somewhat worse network scores. On the whole, these changes then are in the right direction suggesting that key missing theoretical factors were missing. The reduction in the network factors is potentially troubling. This could simply mean we need to reoptimize the parameters that previously led to improvements. Alternatively, and more interestingly, detailed assessment suggests that the changes made to link actors to network is creating projections that are not granular enough. The groups are too large. This suggests we need to better think through the agent-knowledge links and the level of grouping needed

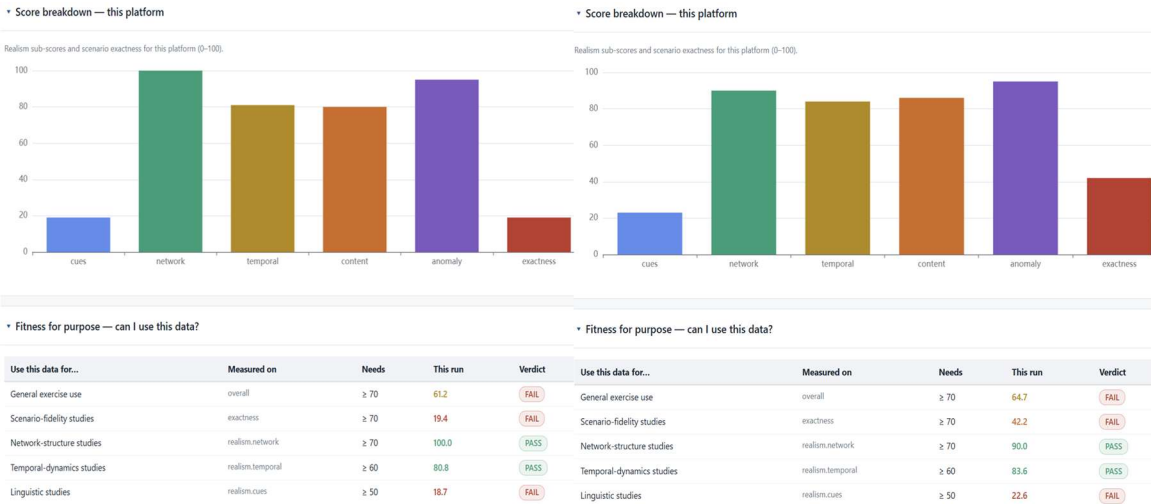


Figure 4. Detailed comparison of generation 3 synthetic data (left) and generation 4 (right).

Table 2 already provides guidance in possible things to do to improve the data realism. Specifically, any of the suggestions not italicized should be developed. In addition, a better approach to finer grained groups is needed. The proposal is to turn to linguistic theory – and extend our agent archetypes with linguistic styles and implement the dynamical limitations due to platform structure and human physiology. The basic idea here is that these micro-affordances of what it means to be human are needed so that the

relation between nodes and attributes are appropriate and acted out in forming and maintaining ties. A second proposal is to employ multiple types of groups (nominal, issue, archetypes ...) to create overlapping and cross-cutting social circles – which will naturally provide opportunities for interaction [14]. The basic idea is that a macro-level topology is needed to enable and constrain the networks (meso-level) with the right structure.

5 Addressing Errors and Limitations

Several strategies can mitigate common errors in synthetic data. Adjusting parameters such as actor distribution, posting frequency, and community structure can improve realism. Incorporating emotional language and refining semantic networks enhances narrative coherence. Additionally, modeling temporal behavior, social cognition, and technological constraints helps align synthetic behavior with real-world dynamics.

The framework categorizes actors into types such as individuals, organizations, and bots, with further differentiation based on roles and communication styles. Linguistic variation plays a critical role in shaping semantic networks, as it reflects differences in stance, tone, and rhetorical purpose. This multi-layered approach ensures that synthetic agents exhibit diverse and realistic communication behaviors.

Achieving true network concomitance requires integrating social cognition, group structures, and behavioral constraints. Actors should be modeled with identities, roles, and linguistic styles, and their interactions should be shaped by shared group memberships and evolving beliefs. Temporal and spatial factors further influence interaction likelihood, contributing to more realistic simulation environments. More to the point, the studies conducted demonstrate that parameter modification is insufficient. They further suggest that to achieve further realism the simulation should be multi-level – with micro level affordances, network formation and evolution at the meso level, and a macro level that sets key topological relations enabled and constrained through group membership. Network concomitance emphasizes the need to model social, semantic, and interaction networks as intertwined systems. Advances in LLM-based synthetic data generation have improved the realism of simulated environments, but challenges remain in achieving full integration. Future work should focus on developing adaptive systems that dynamically capture the co-evolution of actors, content, and interactions, enabling more robust theory building and application in complex social contexts.

6 Conclusion

In conclusion, this paper advances the concept of network concomitance as a foundational principle for theory building in complex socio-informational systems, demonstrating that social, semantic, and knowledge networks must be modeled as co-evolving and interdependent structures. The GhostField framework provides an important step toward operationalizing this perspective by integrating agent-based dynamic-network models with large language models, enabling the generation of synthetic environments that capture both structural patterns and communicative behaviors. The iterative development process reveals that achieving realism is not simply a matter of tuning parameters or prompts, but requires embedding deeper theoretical constructs—such as actor heterogeneity, event salience, and social cognition—into the generative process. These findings underscore that the validity of synthetic data hinges on its ability to reproduce not only observable network features, but also the underlying mechanisms that drive the co-evolution of actors, ideas, and interactions.

At the same time, the study highlights the persistent challenges in aligning structural realism with semantic coherence and behavioral fidelity. While successive generations of synthetic data exhibited improved modularity, community structure, and scenario alignment, deficiencies in narrative diversity, knowledge network formation, and cross-level integration remain. The results demonstrate that network concomitance is difficult to achieve without a multi-level modeling approach that incorporates micro-level cognitive and linguistic constraints, meso-level network dynamics, and macro-level structural and contextual conditions. This suggests that future advances will depend on extending theory to more

explicitly account for attention, emotion, temporal dynamics, and group-based processes, as well as on refining computational architectures to better capture these interactions.

Looking forward, the framework introduced here establishes a robust foundation for the systematic generation and evaluation of synthetic data for research, training, and policy applications. By treating synthetic environments as laboratories for theory development, researchers can iteratively test and refine assumptions about how information ecosystems function under varying conditions. Future work should focus on improving the granularity of group structures, enhancing linguistic and cultural realism, and developing adaptive systems capable of learning from discrepancies between simulated and real-world data. Ultimately, advancing network concomitance as both a theoretical and methodological paradigm will enable more accurate modeling of complex social systems and strengthen our capacity to understand, anticipate, and respond to emergent informational dynamics.

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